Fault Attacks

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Differential Fault Analysis of DES

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  e.g. differential cryptanalysis observes plaintext/ciphertext, timing attacks measure processing time, power attacks measure chip consumption, etc.

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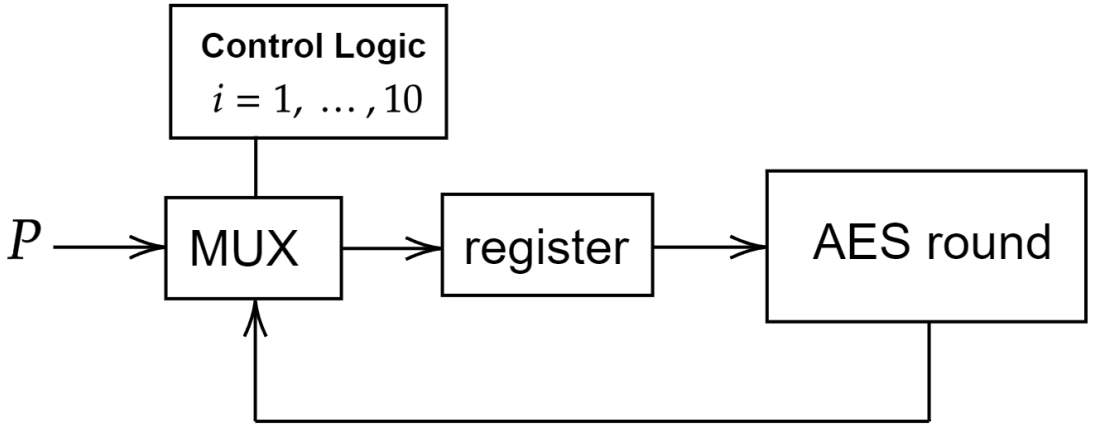
  e.g. differential cryptanalysis observes plaintext/ciphertext, timing attacks measure processing time, power attacks measure chip consumption, etc.
- We will now move to **active** attacks where the adversary does not just observe but can also modify the computation

Introduction

Fault Injection: Stuck-at faults in control logic



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- A control logic circuit keeps track of the round i

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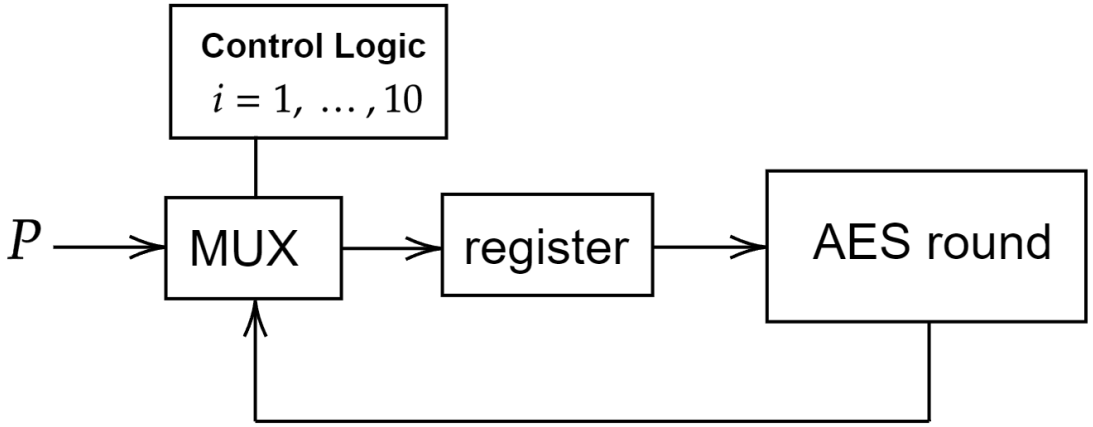

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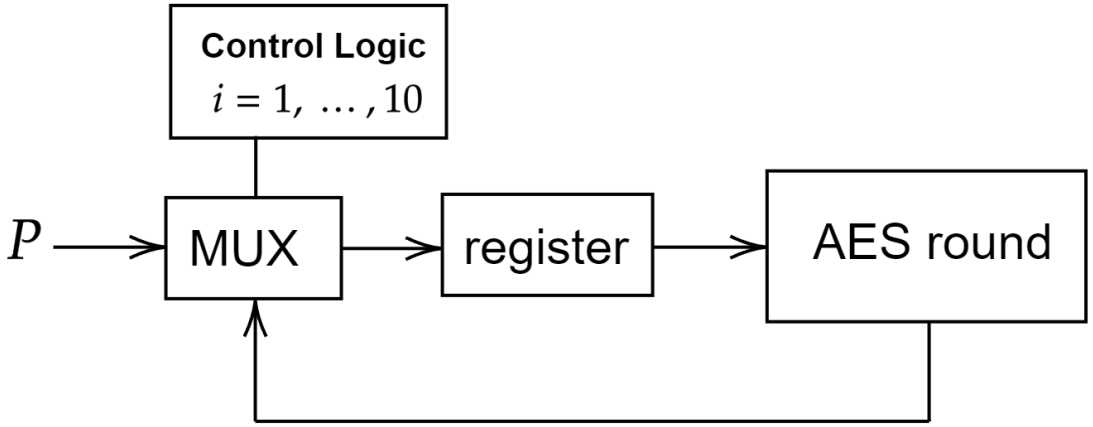

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- A control logic circuit keeps track of the round i
- What would happen if the logic circuit is faulted and gets stuck to $i = 1$?
- We would perform a very weak encryption

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Fault Injection: Instruction Skip

- Assume that a PIN check gets compiled to the following assembly code

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- If the instruction opcode gets altered it could result in an unknown opcode
- The CPU treats unknown opcodes as `nop` instructions
- Thus the jump to the function never happens and the PIN code check gets skipped

Introduction

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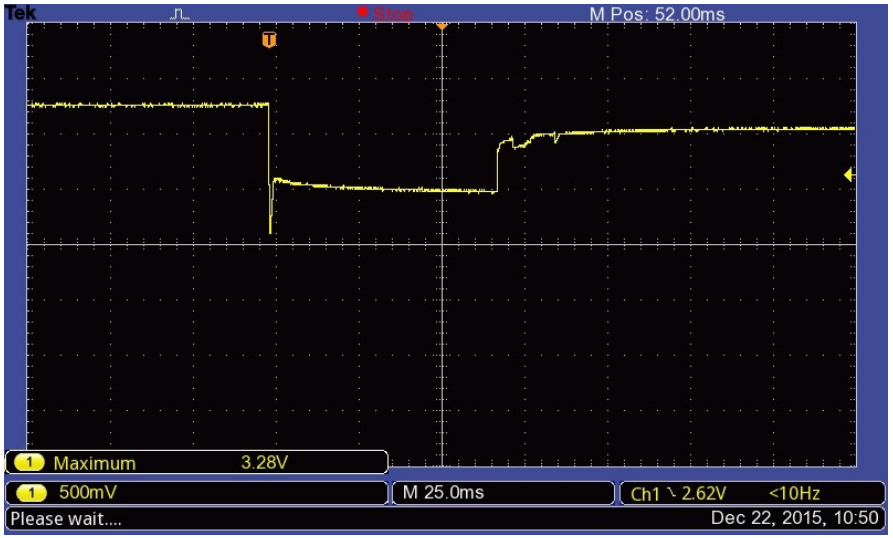
- Decrease or increase the power supply for a very small time

- **Voltage glitching** is one of the earliest fault attacks yet remains simple, effective and applicable to numerous devices. It is a **non-invasive technique** since it does not need device modifications.



- On the downside voltage glitches result often in coarse faults

Introduction

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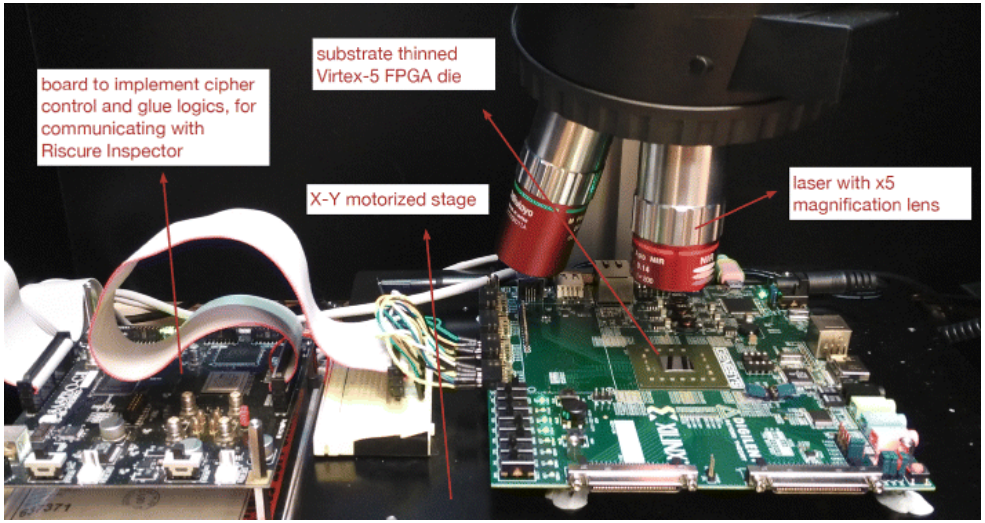
- **Laser-based fault injection** is a more complex process that can result in fine-grained control over the injected fault. It is a **semi-invasive** technique since it typically needs chip decapsulation



- On the downside, achieving precise faults may require a large parameter search effort, trying to locate the exact circuit spot, the right laser intensity, etc.

Introduction

Fault Model:

- Various methods will produce different types of faults, thus many fault categories exist

- **Granularity:** the fault can alter a single-bit or multiple-bits or several bytes
- **Modification:** the faulted value is affected by random bitflips or is now biased according to some statistical distribution. Similarly the faulted value can be stuck-at-zero or stuck-at-one.
- **Control:** The fault can be injected on a large chip region or on a small part of the surface. Likewise the fault can be injected with low and high precision in time
- **Duration:** The fault can be transient (the faulted value reverts), persistent (we need to reset the device to revert the faulted value), permanent (the faulted value cannot be altered)

Differential Fault Analysis of DES

DFA

- Original attack by Biham and Shamir (1997)
- **Goal:** Recover the device's key
- To perform DFA, we encrypt the same plaintext P twice: once in a fault-free manner and once while faulting the encryption algorithm

$$\text{fault-free encryption: } C = enc(P, K)$$

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- The attack exploits the difference between C and C' (i.e. $C \oplus C'$) to obtain information about the secret key K

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- The attack exploits the difference between C and C' (i.e. $C \oplus C'$) to obtain information about the secret key K
- For the attack to work, we must generate several plaintexts P, capture the respective ciphertext pairs (C, C') and use their difference to gradually recover the constant key K

DFA

Fault model and propagation:

- DES has an internal state of 64 bits, split in the left half L and the right half R

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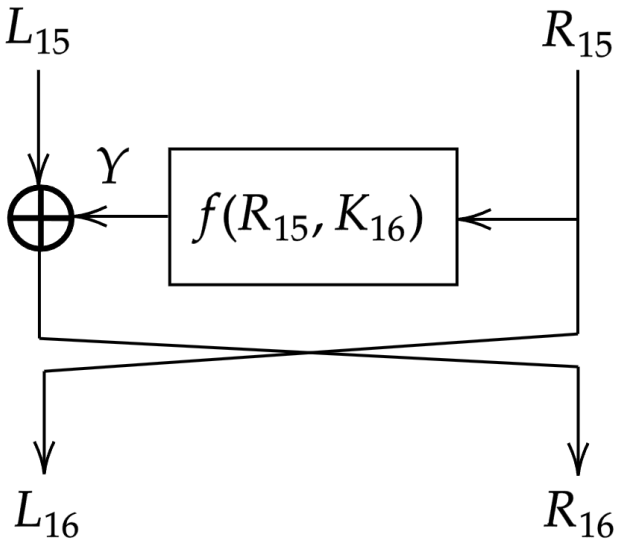


fault-free DES round 16

- In the Feistel construction with round function $f(\cdot)$ it holds that:

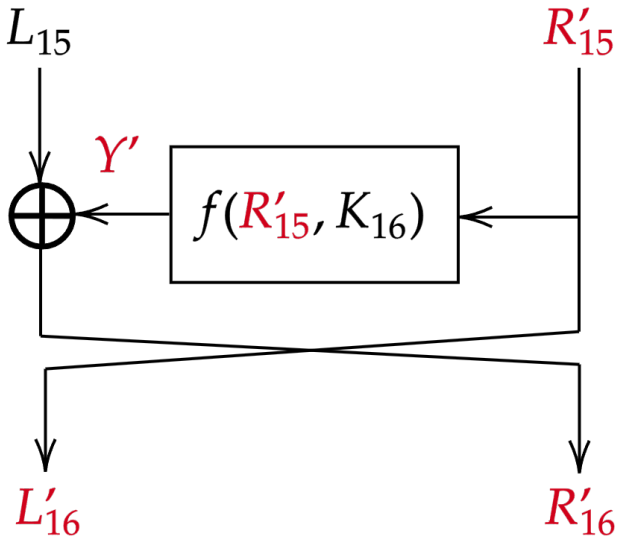
$$L_{16} = R_{15}$$

$$R_{16} = L_{15} \oplus f(R_{15}, K_{16})$$

- To produce the DES ciphertext C we apply the final permutation $FP(\cdot)$ to the 16th round output:

$$C = FP([L_{16}, R_{16}])$$

DFA



faulted DES round 16

- A fault is injected in R_{15} causing bitflips

$$R'_{15} = R_{15} \oplus \epsilon$$

$$\epsilon \in_R \{0,1\}^{32}$$

- The Feistel construction propagates the fault to the output:

$$L'_{16} = R'_{15}$$

$$R'_{16} = L_{15} \oplus f(R'_{15}, K_{16})$$

$$C' = FP([L'_{16}, R'_{16}])$$

- This DFA fault is not very hard to achieve: transient duration (the fault must affect a single DES encryption), not very granular (fault 32 bits of R_{15}), small control (fault R_{15} at any point in point in time before the final round)

DFA

Attack Idea:

- The fault-free and faulted ciphertexts (C, C') are available to the attacker. DFA applies the inverse final permutation $FP(\cdot)$ of DES to compute $[L_{16}, R_{16}]$ and $[L'_{16}, R'_{16}]$ from (C, C')

$$[L_{16}, R_{16}] = FP^{-1}(C), \quad [L'_{16}, R'_{16}] = FP^{-1}(C')$$

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- **Fault differential.** We define the following XOR-difference between the fault-free value R_{16} and the faulted value R'_{16}

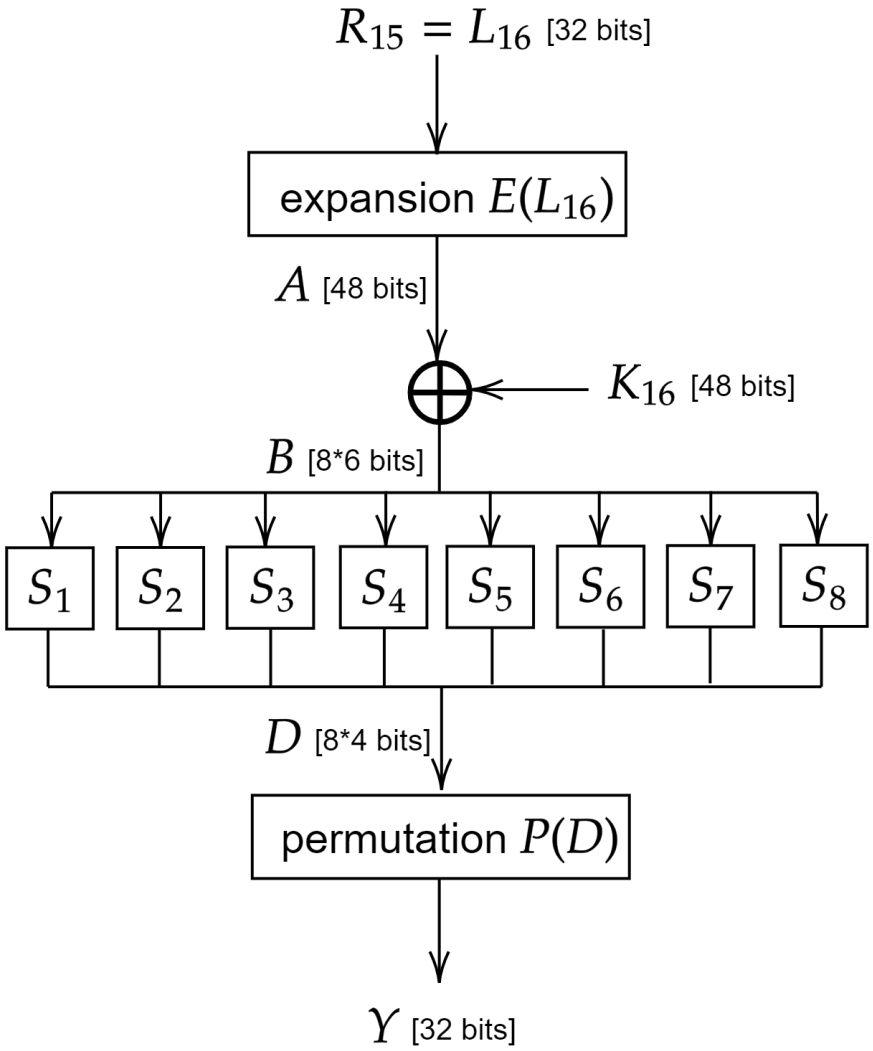
$$\Delta R_{16} \stackrel{\text{def}}{=} R_{16} \oplus R'_{16} = (L_{15} \oplus f(R_{15}, K_{16})) \oplus (L_{15} \oplus f(R'_{15}, K_{16})) =$$

$$f(R_{15}, K_{16}) \oplus f(R'_{15}, K_{16}) = f(L_{16}, K_{16}) \oplus f(L'_{16}, K_{16})$$

- The fault differential ΔR_{16}, together with values L_{16} and L'_{16} will be used to recover information about the round key K_{16}

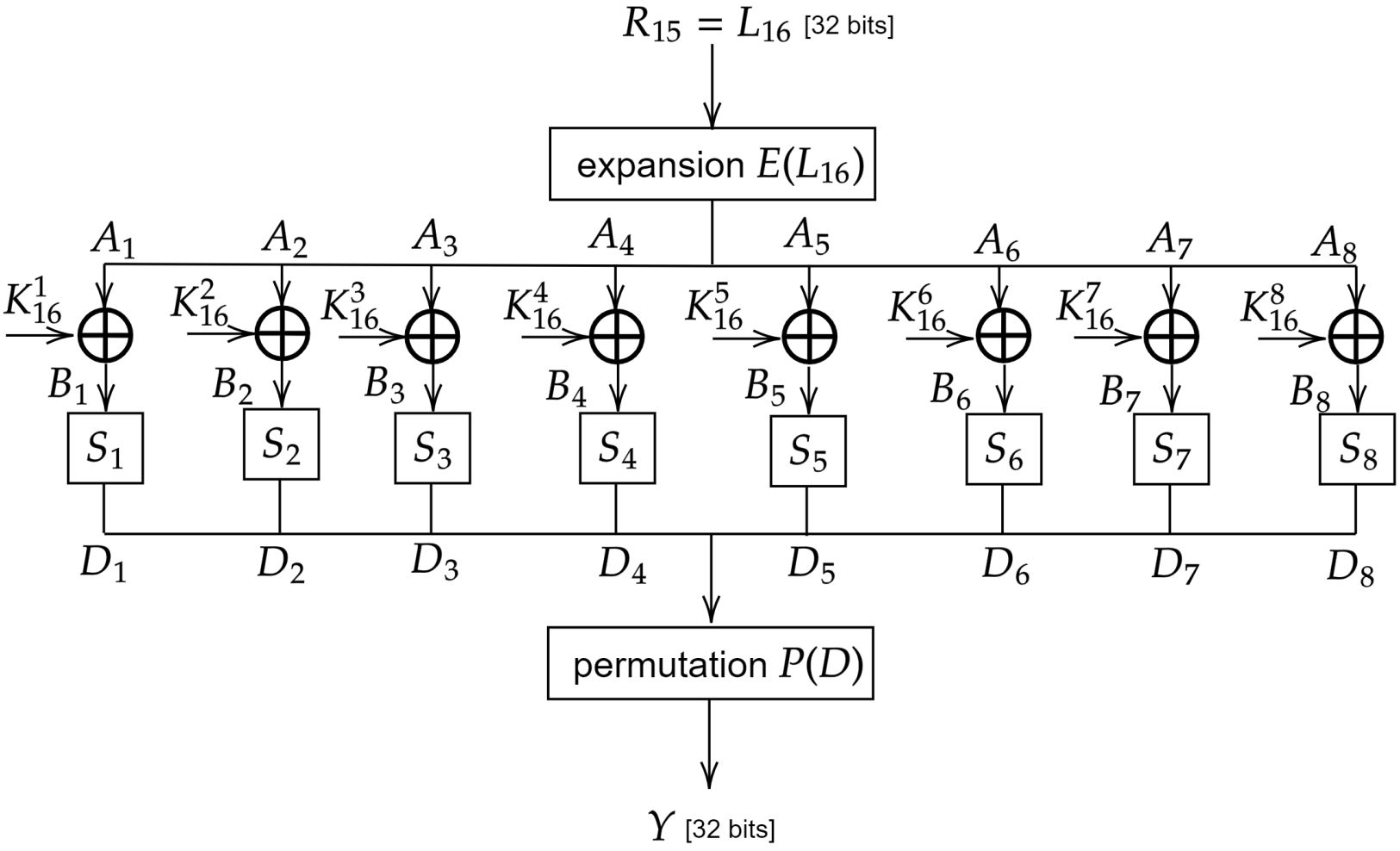
DFA

- Reminder of the DES function $f(\cdot)$ on round 16



DFA

- Divide-and-conquer approach



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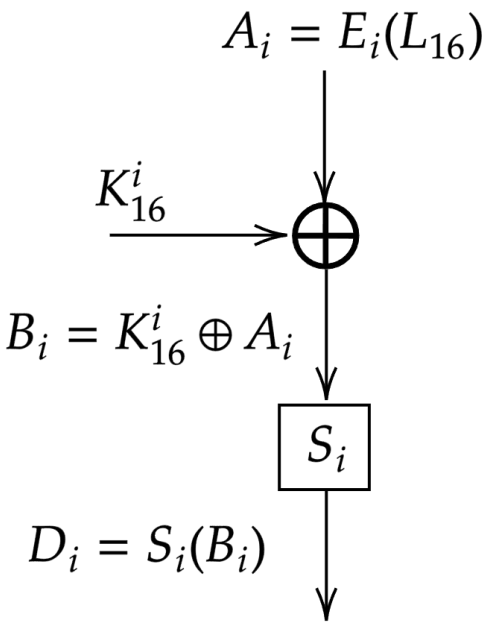


divide-and-conquer

- We isolate the ith sbox $S_i(\cdot)$ of the DES function $f(\cdot)$

$$i = 1, 2, \ldots, 8$$

- The 32-bit input L_{16} gets expanded to the 48-bit value A. We isolate the ith 6-bit part of A and index it as A_i

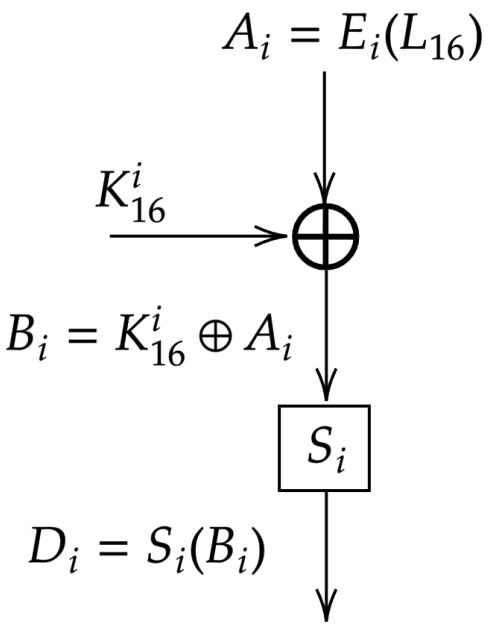
$$A = E(L_{16}), \quad A_i = E_i(L_{16})$$

<u>Notation:</u> $E_i(L_{16})$ reads as: "we compute the 48-bit value $E(L_{16})$ and select the ith 6-bit part of the result"

- The value A_i gets XORed with the 6-bit key part K_{16}^i. The result is the 6-bit value B_i

$$B_i = K_{16}^i \oplus A_i$$

DFA



divide-and-conquer

- The ith sbox of DES $S_i(\cdot)$ is applied to B_i
  The result is the 4-bit value D_i

$$D_i = S_i(B_i)$$

- Putting these steps together we get:

$$D_i = S_i(B_i) \iff$$

$$D_i = S_i(K_{16}^i \oplus A_i) \iff$$

$$D_i = S_i(K_{16}^i \oplus E_i(L_{16}))$$

DFA

- Using the fault differential ΔR_{16} and the structure of $f(\cdot)$ we have:

$$\Delta R_{16} = f(L_{16}, K_{16}) \oplus f(L'_{16}, K_{16}) \iff$$

$$\Delta R_{16} = Y \oplus Y' \iff \Delta R_{16} = P(D) \oplus P(D')$$

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- We apply the inverse permutation $P^{-1}(\cdot)$ to both sides of the equation:

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- We isolate the ith DES sbox:

$$P_i^{-1}(\Delta R_{16}) = D_i \oplus D'_i \iff$$

$$\boxed{P_i^{-1}(\Delta R_{16}) = S_i(E_i(L_{16}) \oplus K_{16}^i) \oplus S_i(E_i(L'_{16}) \oplus K_{16}^i)}$$

Using the **DFA equation** we are able to recover the 6-bit key part K_{16}^i

DFA

Putting the DFA attack together:

1. Acquire faulty ciphertexts: generate n random plaintexts P and repeat the following process. Store the n ciphertext pairs (C, C').

```
for j=1 until n do
    P ←R {0,1}^64
    C = enc(P, K)
    C' = enc↯(P, K)
end
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$$FP^{-1}(C), \quad FP^{-1}(C')$$

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2. Invert the DES final permutation $FP(\cdot)$ for the ciphertext pair (C, C')

$$FP^{-1}(C), \quad FP^{-1}(C')$$

3. Split the pair $(FP^{-1}(C), FP^{-1}(C'))$ to left and right parts

$$[L_{16}, R_{16}] \leftarrow FP^{-1}(C), \quad [L'_{16}, R'_{16}] \leftarrow FP^{-1}(C')$$

DFA

4. Compute the fault differential ΔR_{16} and apply the inverse permutation $P^{-1}(\cdot)$

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5. Compute the expansion $E(\cdot)$ of values L_{16} and L'_{16}

$$E(L_{16}), \quad E(L'_{16})$$

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$$E(L_{16}), \quad E(L'_{16})$$

– The DES function $f(\cdot)$ consists of 8 sboxes $S_i(\cdot)$ and operates on 8 6-bit key parts K^i_{16}, where $i = 1, 2, \ldots, 8$

– We try to recover the 1st 6-bit key part (K^1_{16}) i.e. we focus on sbox $i = 1$. The same process will be repeated for all sboxes (8 in total).

DFA

7. For all possible values of K_{16}^1 i.e. for $k \in \{0, 1, 2, \ldots, 2^6 - 1\}$
   7.1 Isolate the values related to sbox S_1

$$P_1^{-1}(\Delta R_{16}) \xleftarrow{i=1} P_1^{-1}(\Delta R_{16}), \quad E_1(L_{16}) \xleftarrow{i=1} E(L_{16}), \quad E_1(L'_{16}) \xleftarrow{i=1} E(L'_{16})$$

   7.2 Construct the DFA equation

$$P_1^{-1}(\Delta R_{16}) = S_1(E_1(L_{16}) \oplus k) \oplus S_1(E_1(L'_{16}) \oplus k)$$

   7.3 If the DFA equation holds, then k is a valid candidate for K_{16}^1
   If it does not, then we discard k

DFA

DFA on DES round 16:

```
for j = 1 until n do
    (C, C') ← jth ciphertext pair
    -compute FP^{-1}(C) , FP^{-1}(C')
    -split [L_16, R_16] ← FP^{-1}(C') and [L'_16, R'_16] ← FP^{-1}(C')
    -compute P^{-1}(ΔR_16), E(L_16) , E(L'_16)
    for i = 1 until 8 do
        candidates = ∅
        for k = 0 until 2^6 − 1 do
            P_i^{-1}(ΔR_16) ← P^{-1}(ΔR_16),  E_i(L_16) ← E(L_16),  E_i(L'_16) ← E(L'_16)
            check = P_i^{-1}(ΔR_16) == S_i(E_i(L_16) ⊕ k) ⊕ S_i(E_i(L'_16) ⊕ k)
            if  check then
                candidates = candidates ∪ k
            end
        end
        if j==1 then
            K_16^i = candidates
        else
            K_16^i = K_16^i ∩ candidates
        end
    end
end
```

DFA

Final notes on DFA:

- Very strong attack that requires only a small amount of fault injections and a fairly realistic fault model
- Requires observable ciphertext and the ability to keep the plaintext constant for a fault-free and a faulted encryption
- DFA has many variants: various fault models can be injected on different DES rounds or other ciphers

DFA

Countermeasures:

- **Redundancy.** Run the encryption with the same plaintext input more than once and compare the ciphertexts. If they match then no fault was injected.
- **Error detection/correction.** Enhance the cipher with a scheme that detects and/or corrects tampering of internal values during computation
- **Sensors.** Implement sensors on the device that detect e.g. voltage fluctuations or laser injections.
- Still, ciphertext comparison can also be faulted, certain faults (ineffective faults) can bypass redundancy/error detection and sensors may accidentally render the device useless